

Artificial Intelligence

An Advanced Course About Foundations



Probabilistic Reasoning: Representation & Inference

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Probability Space

Probability Space (*preliminary definition*)

■ **Probability space**

A triple $\langle W, \Sigma, P \rangle$

Possible worlds
(a.k.a. Sample
Space)

Event Space
(a collection of
subsets over W)

Probability Measure

$$P : \Sigma \rightarrow [0, 1]$$

*The intuitive definition is simple enough,
its mathematical translation ... not so much*

Event Space: *a collection of subsets of possible worlds*

■ Boolean algebra

A non-empty collection of subsets Σ of a set W such that:

- 1) $A, B \in \Sigma \implies A \cup B \in \Sigma$
- 2) $A \in \Sigma \implies A^c \in \Sigma$
- 3) $\emptyset \in \Sigma$

Corollary:

The sets $\emptyset$ and W belong to any Boolean algebra generated on W
 Σ is also closed under binary intersection

■ σ -algebra

A non-empty collection of subsets Σ of a set W such that:

- 1) $A_k \in \Sigma, \forall k \in \mathbb{N}^+ \implies (\bigcup_{k=1}^{\infty} A_k) \in \Sigma$
- 2) $A \in \Sigma \implies A^c \in \Sigma$
- 3) $\emptyset \in \Sigma$

*This is a stronger requirement:
closeness under countable union
Hence a σ -algebra is a boolean algebra
but not vice-versa*

Corollary:

The sets $\emptyset$ and W belong to any σ -algebra generated on W
 Σ is also closed under countable intersection

Probability Measure

- σ -algebra (*Event Space*)

A non-empty collection of subsets Σ of a set W such that:

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- Probability measure over a σ -algebra (i.e., over the events)

A function $P : \Sigma \rightarrow [0, 1]$

i.e. P assigns a measure (i.e. a real number)
to each elements of a σ -algebra Σ of subsets of W

- 1) $\forall A \in \Sigma, P(A) \geq 0$
- 2) $A_1, A_2 \in \Sigma$ are disjoint $\implies P(A_1 \cup A_2) = P(A_1) + P(A_2)$
 $A_k \in \Sigma, \forall k \in \mathbb{N}^+$ are all disjoint $\implies P(\bigcup_{k=1}^{\infty} A_k) = \sum_{k=1}^{\infty} P(A_k)$
- 3) $P(\emptyset) = 0$
- 4) $P(A^c) = 1 - P(A)$ (which implies $P(W) = 1$)

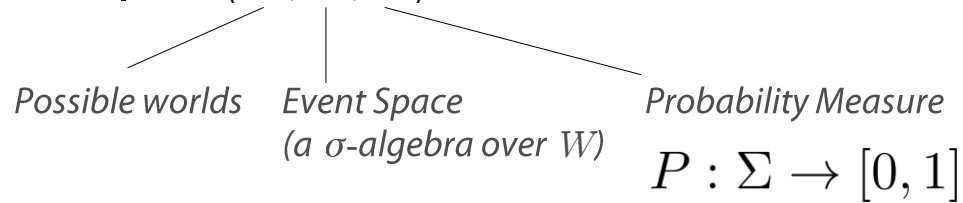
Finite additivity

Countably infinite additivity

Probability Space

■ **Probability space**

A triple $\langle W, \Sigma, P \rangle$



Why bothering so much with these (very) technical definitions?

■ **Rationale** *(just a few hints)*

Closure w.r.t. *countable unions* of a σ -algebra (as well as *countable additivity* of P) is required for dealing with *infinite sequences of events*

A σ -algebra is included in the *power set* of W (i.e., the collection of all its possible subsets): requiring closure on countable union is a *restriction*, to ensure measurability

(see the so-called Banach-Tarski paradox for counterexamples)

An Aside: Probability is Systemic, not Compositional

In general:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

It follows from the additivity property

If $A \cap B = \emptyset$ then events A and B are disjoint

$$P(A \cup B) = P(A) + P(B)$$

(*) Note that $A \cap B = \emptyset \implies P(A \cap B) = 0$

but not vice-versa: as an event can have zero probability without being empty

(**) Unlike in propositional logic, knowing the basic values $P(A)$ and $P(B)$ is not sufficient for determining the value of composite events like $P(A \cup B)$

Namely, probability is not *compositional* ...

Discrete Probability

Studying basic properties in a **finitary* setting

A simpler setting that allows a more intuitive definition of fundamental properties

- Finite event space

Σ is a finite collection of subsets of W

In such setting:

σ -algebra $\equiv$ Boolean algebra

*Events could also be defined via propositional logic
(à la de Finetti, 1937)*

- Finitely additive probability measure

Just summations, no integrals

Computability will be always guaranteed

Random Variables*

(*) In a finitary setting

Partitions, random variables*

- Partition

A finite collection of n *disjoint* subsets $A_i \subseteq W$ such that

$$\bigcup_{i=1}^n A_i = W$$

A σ -algebra (*Boolean algebra, in fact*) can be generated from a *partition* by adding its closure under *union* plus the empty set $\emptyset$

Partitions, random variables*

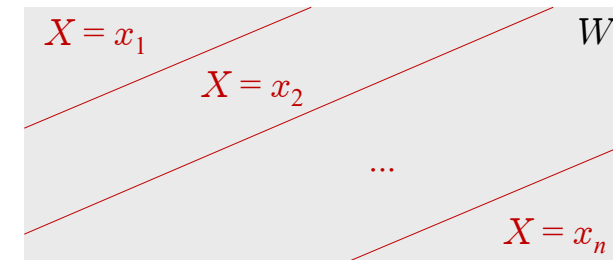
(*) In a finitary setting

■ Random Variable (i.e. away to define a σ -algebra)

Let X be a variable having a finite set of possible values $\{x_1, x_2, \dots, x_n\}$

In each possible world, the variable X is assigned one specific value x_i

- The set of all possible assignments $\{X = x_1, X = x_2, \dots, X = x_n\}$ defines a partition over W
- A σ -algebra can be obtained by taking the closure of the partition under union plus the empty set
- $X = x_i$ defines an event (i.e. a subset of W)
- $X = x_i$ and $X = x_j$ are disjoint events, whenever $i \neq j$



Partitions, random variables*

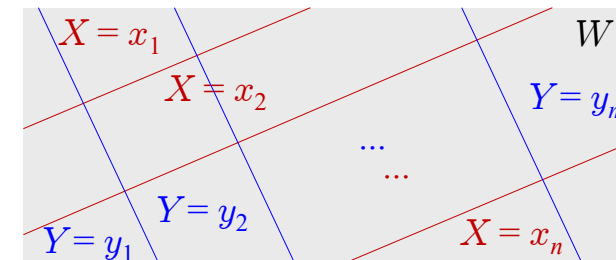
(*) In a finitary setting

Multiple random variables

In practice, in a probabilistic representation, there will be multiple random variables

The intersection of two or more σ -algebras is still a σ -algebra

Together, a collection of random variables defines a partition of W



Random variables, distribution, joint distribution*

(*) In a finitary setting

■ Probability Distribution

for one random variable

It is a function that associates a value in $[0, 1]$ to each value assignment

$$P(X = x) \in [0, 1], \quad \forall x \in \mathcal{X} \quad \text{--- The set of possible values for variable } X$$

Given that any two $x_i, x_j \in \mathcal{X}$, $i \neq j$ define two disjoint events

$$P(X = x_i \cup X = x_j) = P(X = x_i) + P(X = x_j)$$

and

$$\sum_{x \in \mathcal{X}} P(X = x) = 1$$

Also written as (*shorthand notation*)

$$\sum_X P(X) = 1$$

Random variables, distribution, joint distribution*

(*) In a finitary setting

■ Joint Probability Distribution

for a set of random variables (for example X , Y and Z)

It is a function that associates a value in $[0, 1]$ to each combination of value assignments

$$P(X = x, Y = y, Z = z) \in [0, 1], \quad \forall x \in \mathcal{X}, y \in \mathcal{Y}, z \in \mathcal{Z}$$

Given that X , Y and Z induce a *combined partition* over W

$$\sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} \sum_{z \in \mathcal{Z}} P(X = x, Y = y, Z = z) = 1$$

Also written as (*shorthand notation*)

$$\sum_X \sum_Y \sum_Z P(X, Y, Z) = 1$$

In joint probability notation, the order of variables is interchangeable

Events and functions: notation*

(*) In a finitary setting

■ Random variables, events and σ -algebras

Sometime the shorthand notation can be confusing

$$P(X)$$

This is the probability measure over the σ -algebra generated by the random variable X

$$P(X = x) \qquad P(x) \quad (\text{shorthand notation})$$

This the probability (i.e. a value in $[0,1]$) associated to the event $X = x$

$$P(X, Y = y) \qquad P(X, y) \quad (\text{shorthand notation})$$

This is the probability measure over the σ -algebra generated by the random variable X in the subspace of W corresponding to the event $Y = y$

Fundamental Operations*

(*) In a finitary setting

Marginalization*

(*) In a finitary setting

"Removing" a random variable from a joint distribution

Given a joint probability distribution

$$P(X, Y)$$

The marginal probability $P(X)$ is obtained via a summation:

$$P(X) := \sum_{y \in \mathcal{Y}} P(X, Y = y)$$

A marginal probability can be a joint probability as well ...

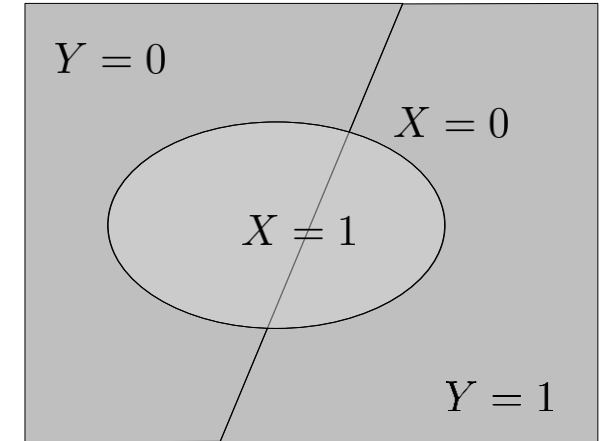
$$P(X, Y) := \sum_{z \in \mathcal{Z}} P(X, Y, Z = z)$$

Shorthand notation, values omitted

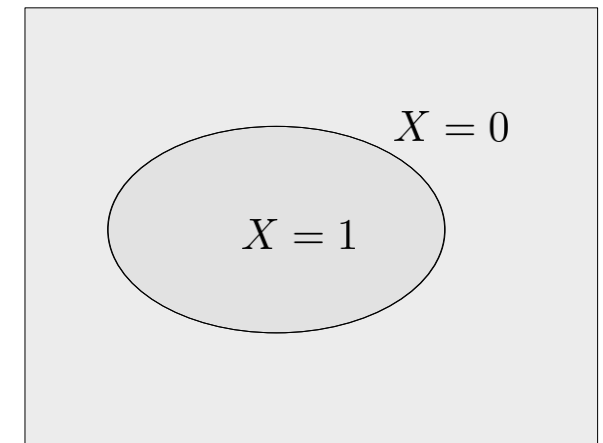
$$P(X) = \sum_Y P(X, Y)$$

$$P(X, Y) = \sum_Z P(X, Y, Z)$$

$$P(X, Y)$$



$$P(X)$$



Conditionalization*

(*) In a finitary setting

Exploring 'what if' something becomes known

Given a joint probability distribution

$$P(X, Y)$$

The conditional probability $P(X \mid Y = y)$ is defined as:

$$P(X \mid Y = y) := \frac{P(X, Y = y)}{P(Y = y)}$$

A conditional probability can be a joint probability as well ...

$$P(X, Y \mid Z = z) := \frac{P(X, Y, Z = z)}{P(Z = z)}$$

Conditional probability, more general (**not** shorthand) notation :

$$P(X \mid Y) := \frac{P(X, Y)}{P(Y)}$$

Conditional probabilities for the whole σ -algebra generated by Y
(it represents a family of probability measures)

Conditionalization*

(*) In a finitary setting

Exploring 'what if' something becomes known

Conditional probability, more general notation:

$$P(X \mid Y) := \frac{P(X, Y)}{P(Y)}$$

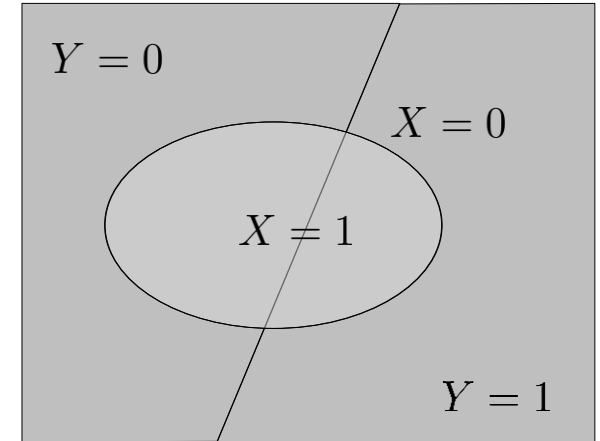
Example: assume both variables are *binary* $X, Y \in \{0, 1\}$

$$P(X \mid Y = 0) := \frac{P(X, Y = 0)}{P(Y = 0)}$$

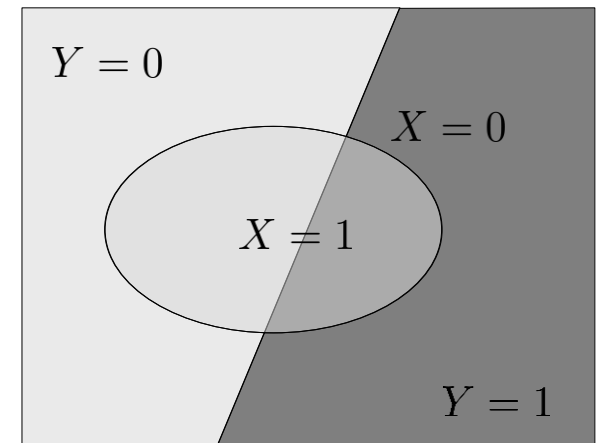
$$P(X \mid Y = 1) := \frac{P(X, Y = 1)}{P(Y = 1)}$$

Each value of the conditioning variable defines a distinct event (sub)space

$P(X, Y)$



$P(X \mid Y = 0)$



Chain Rule

Starting point: a joint distribution

$$P(X, Y, Z)$$

conditioning on X :

$$P(Y, Z|X) = \frac{P(X, Y, Z)}{P(X)}$$

which implies:

$$P(X, Y, Z) = P(X) P(Y, Z|X)$$

then, conditioning on Y :

$$P(Z|X, Y) = \frac{P(Y, Z|X)}{P(Y|X)}$$

which implies:

$$P(Y, Z|X) = P(Y|X) P(Z|X, Y)$$

hence and finally:

$$P(X, Y, Z) = P(X) P(Y|X) P(Z|X, Y)$$

Univariate factorization of the joint distribution

Note that we used $X \rightarrow Y \rightarrow Z$: any other sequence would generate a valid factorization

The Bayes' Theorem

Bayes' Theorem* (T. Bayes, 1764)

(*) In a finitary setting

■ Definition

A relation between conditional and marginal probabilities

$$P(X|Y) = \frac{P(Y|X)P(X)}{P(Y)}$$

$P(Y|X)$ is also called the *likelihood* $L(X|Y)$



The theorem follows from the definition of conditional probability (*chain rule*)

$$P(X, Y) = P(X|Y)P(Y) = P(Y|X)P(X)$$

Furthermore, given the definition of marginalization:

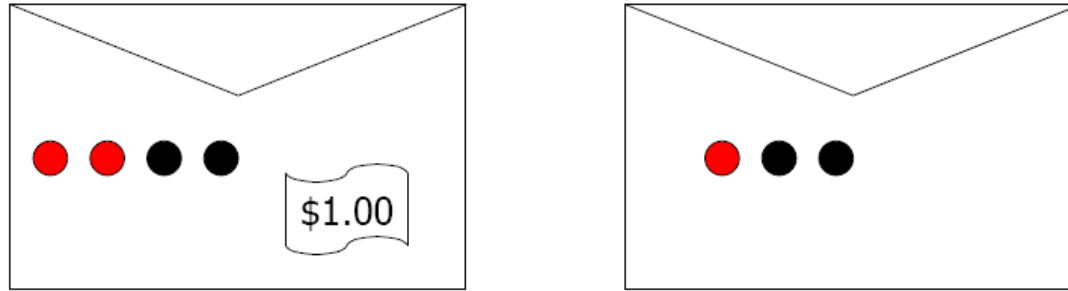
$$P(Y) = \sum_X P(X, Y) = \sum_X P(Y|X)P(X)$$

Also called
'law of total probability'

it follows an alternative formulation of the Bayes' theorem:

$$P(X|Y) = \frac{P(Y|X)P(X)}{\sum_X P(Y|X)P(X)}$$

Example: information and bets



- Two envelopes, only one is extracted

One envelope contains two red tokens and two black tokens, it is worth \$1.00

One envelope contains one red token and two black tokens, it is valueless

The envelope has been extracted.

Before posing you bet, you are allowed to extract one token from it

- a) The token is black. How much do you bet ?
- b) The token is red. How much do you bet ?

Purpose: showing that Bayes' Theorem makes the representation easier

Inference (without *learning*)

Probabilistic Inference* (*general structure*)

(*) In a finitary setting

■ General structure of probabilistic inference problems

The starting point is a fully-specified joint probability distribution

$$P(X_1, X_2, \dots, X_n)$$

In an *inference* problem, the set of random variables $\{X_1, X_2, \dots, X_n\}$ is divided into three categories:

- 1) Observed variables $\{X_o\}$ *having a definite (and supposedly known) value*
- 2) Irrelevant variables $\{X_i\}$ *which are neither observed nor directly part of the answer*
- 3) Relevant variables $\{X_r\}$ *which are part of the answer we seek*

Performing inference means computing:

$$P(\{X_r\} \mid \{X_o\}) = \sum_{\{X_i\}} P(\{X_r\}, \{X_i\} \mid \{X_o\}) = \sum_{\{X_i\}} \frac{P(\{X_r\}, \{X_i\})}{P(\{X_o\})}$$

- “Decidability” (“computability”) is guaranteed (*in a finitary setting)

Given that the joint probability distribution is completely specified

- Computational efficiency can be a problem

The number of value combinations to be considered for the computation grows exponentially with the number n of random variables

Independence

Independence, conditional independence

■ Independence (also *marginal independence*)

Two variables are independent

iff their joint probability can be factorized into the product of *marginals*

$$\begin{aligned} \langle X \perp Y \rangle &\Leftrightarrow P(X, Y) = P(X)P(Y) \Leftrightarrow \langle Y \perp X \rangle \\ &\Rightarrow P(X|Y) = \frac{P(X, Y)}{P(Y)} = \frac{P(X)P(Y)}{P(Y)} = P(X) \end{aligned}$$

■ Conditional independence

Two variables are conditionally independent given a third variable,

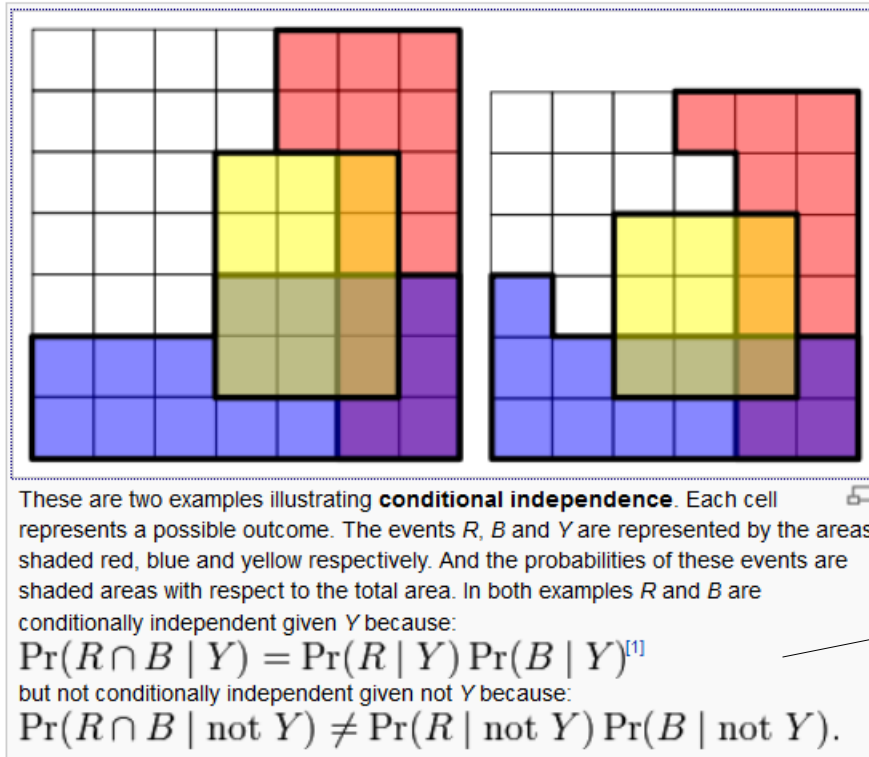
iff their joint conditional probability can be factorized into the product of *conditional marginals*

$$\begin{aligned} \langle X \perp Y | Z \rangle &\Leftrightarrow P(X, Y | Z) = P(X | Z)P(Y | Z) \Leftrightarrow \langle Y \perp X | Z \rangle \\ &\Rightarrow P(X | Y, Z) = \frac{P(X, Y | Z)}{P(Y | Z)} = \frac{P(X | Z)P(Y | Z)}{P(Y | Z)} = P(X | Z) \end{aligned}$$

CAUTION: the two forms of independence are distinct!

$$\langle X \perp Y \rangle \not\Leftrightarrow \langle X \perp Y | Z \rangle \quad \langle X \perp Y | Z \rangle \not\Leftrightarrow \langle X \perp Y \rangle$$

Independence, conditional independence



[from Wikipedia, "Conditional Independence"]

R , B and Y here are subsets, i.e. events,
not random variables

The example above shows that (marginal or conditional) independence of two specific events does NOT imply (marginal or conditional) independence of the whole σ -algebras

Independence is assumed to be between σ -algebras, not just specific combinations of events

Continuous Random Variables

Continuous random variables (hints)

Although intuitively similar, dealing with continuous random variables is technically difficult

Consider a **continuous** random variable $X \in \mathcal{X}$ — A continuous value space:
e.g. the real interval $[0, 1]$
or even $[-\infty, +\infty]$

$X = x$ does not describe an event in a continuous setting

For technical reasons (of *measurability*), a point must have probability zero

Events need to be *subsets*, or better, intervals:

$$X \leq a, X \leq b, \quad a < X \leq b \quad \text{— Assuming } a < b$$

Probability measures these *subsets*

$$P(X \leq b) = P(X \leq a) + P(a < X \leq b) \quad \text{— These two events are disjoint}$$

$$P(a < X \leq b) = P(X \leq b) - P(X \leq a)$$

$$P_X(x) = P(X \leq x) \quad \text{— Shorthand notation}$$

Density and Cumulative Distribution

■ Probability Density Function (pdf)

It is defined as the derivative

$$p_X(x) := \frac{dP_X(x)}{dx}$$

provided that it exists everywhere in $\mathcal{X}$

■ Probability Measure as Cumulative Distribution Function (CDF)

Cumulative Distribution Function (CDF)

$$P(a < X \leq b) := \int_a^b p_X(x) dx$$

probability density function (pdf)

As a probability measure, it must integrate to unity

$$\int_{x \in \mathcal{X}} p_X(x) dx = 1$$

Note that $p(x)$ may well be above 1 (its integral over the value space $\mathcal{X}$ will be equal to one)

Expectation and Variance

Expected Value of a Random Variable

(also: *Expectation*)

Basic definition

(*) In a finitary setting

$$\mathbb{E}_X[X] := \sum_{x \in \mathcal{X}} x P(X = x)$$

Shorthand notation*

$$\mathbb{E}[X] := \sum_{x \in \mathcal{X}} x P(x)$$

Continuous case

$$\mathbb{E}[X] := \int_{x \in \mathcal{X}} x p_X(x) dx$$

Also denoted as:
 μ_X

Expectation is a linear operator

$$\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]$$

$$\mathbb{E}[cX] = c\mathbb{E}[X]$$

Conditional expectation*

$$\mathbb{E}_X[X|Y = y] = \mathbb{E}[X|Y = y] := \sum_{x \in \mathcal{X}} x P(X = x|Y = y)$$

Variance of a random variable

Basic definition

$$\text{Var}(X) := \mathbb{E}_X[(X - \mathbb{E}_X[X])^2] = \mathbb{E}_X[(X - \mu_X)^2]$$

where*:

(*) In a finitary setting

$$\text{Var}(X) := \sum_{x \in \mathcal{X}} P(X = x) (x - \mu)^2$$

Also denoted as: σ_X^2

$$\sigma_X := \sqrt{\text{Var}(X)}$$

Standard Deviation

Variance is not a linear operator

Variance lemma

$$\text{Var}(X) = \mathbb{E}[(X - \mu_X)^2] = \mathbb{E}[X^2] - 2\mu_X \mathbb{E}[X] + \mu_X^2$$

$$= \mathbb{E}[X^2] - 2\mu_X^2 + \mu_X^2 = \mathbb{E}[X^2] - \mu_X^2$$

$$\mathbb{E}[X^2] = \mu_X^2 + \sigma_X^2$$

Conditional Variance

$$\text{Var}(X|Y = y) := \mathbb{E}_X[(X - \mathbb{E}_X[X|Y = y])^2 | Y = y]$$